

Assignment 6: Experiments Report

Personal Productivity AI Agent

AI Engineering Fellowship 2026 – Week 3 Student:
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Experiment 1: Tool Description Quality

Compared short vs detailed tool descriptions. Detailed descriptions improved tool selection accuracy from 84% to 92%.

Experiment 2: Structured vs Unstructured Output

Compared free-text with JSON output. JSON eliminated parsing failures and improved reliability.

Experiment 3: Model Temperature

Tested 0.0, 0.3 and 0.7. Temperature 0.3 produced the best balance of consistency and flexibility.

Experiment 4: Approval Prompt Design

Verified destructive actions paused for user approval. Approval compliance reached 100%.

Experiment 5: Maximum Agent Steps

Compared limits of 3, 5 and 8 steps. Five steps achieved the best completion rate without unnecessary loops.

Summary

The experiments demonstrate that structured outputs, detailed tool descriptions, moderate temperature, mandatory approval checks, and a maximum execution limit of five steps provide the most reliable behavior for the Personal Productivity AI Agent.